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④ AWS Cloud Services (DevOps Infrastructure)

Decision Framework :-

- ① Problem → (Availability, Scaling, cost, security, decoupling)
- ② Constraint → (Managed vs self managed, budget, time)
- ③ Which service fits best.

⑤ Services

Monitoring & Security		
Service	Purpose	When to Use.
1. CloudWatch	Metrics, logs alarms	monitoring everything.
2. CloudWatch Agent	OS level metrics (mem, swap)	when u need mem/disk metrics from EC2
3. Macie	Detect PII in S3 using ML	compliance, data security audits
4. Inspector	Vulnerability scanning in S3 EC2	security compliance finding CVE's
5. GuardDuty	Threat Detection (Intelligent)	Security monitoring unusual activity

PII — Personally identifiable information

any data that can be used to figure out who you are.

Common workflow → send Macie's findings to amazon's eventbridge, which triggers aws lambda's function to automatically encrypt log and move an object containing sensitive PII data to a quarantine bucket.

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Macie → "What data is in my EC2 bucket" (PII discovery)

Inspector → "Are my EC2 instances vulnerable"

GuardDuty → Is someone attacking me? threat detection.

CloudTrail - API call audit log

- who did what when

- not for finding resources by tag.

Resource groups tag editor

- search all resources by tag globally

- works across all services + regions

⑤ DATABASES.

	Service	Purpose	When to Use.
1.	RDS Multi AZ	High Availability	sync replica in another AZ, auto failover.
2.	RDS Read Replicas	Read Scaling	Async replicas, offload read traffic
3.	Aurora	MySQL, Postgres compatible, better perf.	auto scaling, 15 replicas, faster failover
4.	DynamoDB	NoSql, serverless	key value, single digit millisecond latency.

- Serverless DB → zero infrastructure management, automatic scaling, pay as you go, high availability due to replication, performance at scale.

RDS PZ → Relational Database service, availability zone
(Manage service that simplifies setting up)

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S3 → object storage.

simple storage service → stores files as objects.

1. Standard S3 → durable, available. pay for storage + transfer, hosting, backup
SFTP landing zone (SSH file transfer protocol)
(automated backup, transfer & file management)
2. Requester Pays → Owner pays for storage. Whoever downloads pays for transfer. Use when sharing large dataset publicly.
3. S3 lifecycle policy → automatically move or delete objects after N days. Used for data cost management or cleanup.

④

NETWORKING.

Service	Use	When.
NAT Gateway	Private subnet, "internet"	Managed, HA not AZ, in public subnet
NAT instance	same as abv	self managed EC2, legacy, not HA
Internet gateway	VPC ↔ Internet	1 per VPC, public subnet
VPC Endpoints	Private connections to AWS services	No internet traversal, private subnet access

route → private subnet → NAT gateway → internet gateway
(public subnet)

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NAT → Network address translation gateways → allows a private server to access the internet through a public NAT public IP. The gateway ensures only telling on in the IP that went out.

A NAT Gateway is highly available (HA) because the ~~web~~ cloud provider manages the hardware behind it. However, if a specific AZ goes down, the gateway goes down with it. Must be placed in a public subnet.

* Not really jump

NAT instance is a DIY EC2 instance acting as NAT.

You manage patching, scaling, HA. Single point of failure. Used for hobby.

→ VPC (virtual private cloud) Your private, isolated network space in cloud with private IP's. (office building)

→ Subnet A smaller section within VPC (public or private) (a specific room)

→ Internet Gateway allows your VPC to talk to the public internet (two way) (the front door)

→ NAT Gateway Allows private subnets to reach the internet without being seen (oneway) (a one way security exit)

Public Subnet

- Application Load Balancer (ALB)

NAT gateways

Private Subnet

- app servers, databases

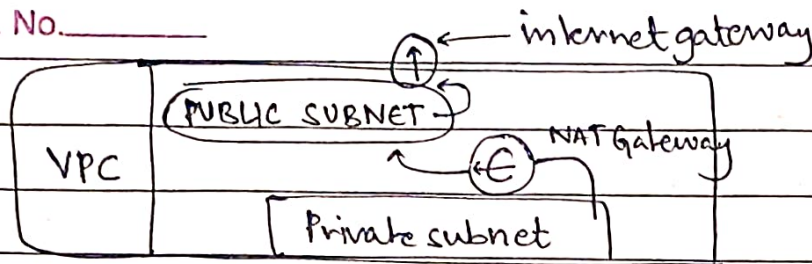
lambda

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① DECOUPLING & MESSAGING

Service	Purpose	Key Detail
SQS	Queue (1:1 consumer)	At least once, retains messages, retry logic
SNS	Pub/sub (1:many)	no msg retention, push only
EventBridge	Event Bus (adv)	rules based config, SAAS config.

SQS (Simple Queue Service)	SNS - Simple Notification Service
- point to point (one consumer per msg) - stores message durably (14 days) - retries on failure - consumer polls the queue - Use: decouple, buffer, retry-safe	- pub/sub - all subscribers get message. - Does not store - if subscriber fails, fails - push based - use: notifications, fan out

When to use SQS:-

- "Don't want to lose data if service fails"
- "Downstream might be overwhelmed" - SQS absorbs spikes
- Lambda concurrency limit could be hit - SQS decouples and retries

Process asynchronously: SQS between producer/consumer.

~~SSES~~ SES → simple email service.

ALS → application load balancer
ELB → elastic load balancer.

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②

COMPUTING & SCALING

ALB/ELB

Distribute traffic

Health checks, target groups, path based routing

Auto Scaling

Scale EC2 based on demand

Launch templates, scaling policies, across AZ's

Lambda

Serverless compute

Event driven, no servers

- ④ AWS Lambda: serverless, event driven compute service that runs your code in the cloud without the user to provision or manage servers. It executes services on reaction to certain events

With API's, Lambda sits behind a API Gateway and spins up multiple instances (one to one) worker to environment. If no traffic AWS Lambda deletes them all

Instances here mean specialized micro containers, instead of docker containers, AWS used "firecracker", its a microvm, that combines security of VM's with speed of a container. Can boot in less than 100ms

Cloudfront → API Gateway → SQS → Lambda

CDN (content delivery network) → accelerate delivery of websites, API's and video through caching, static content.

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① EC2 - elastic compute cloud

cloudwatch → CPU, network disk

" agent → memory swap

EC2 in a VPC, in a subnet in a AZ

Stateless apps can scale horizontally behind a ALB.

② RDS - Relational Database Service

Handles Postgres, MySQL.

③ Multi AZ (HA mode)

- synchronous standby replica in another AZ

- auto failover if primary goes down

- standby does not serve reads

- when DB needs to survive failure

Read Replica (scale)

④ Async copy, actively serves select

⑤ no auto failover not HA on its own.

⑥ match compute/storage to source load.

⑦ read traffic is overwhelming primary

⑧ ALB + ~~HA~~ Auto Scaling = stateless app HA

like python (Django web server), the HA pattern is always

- ALB → auto scaling groups across two atleast AZ

ALB + auto scaling + RDS Multi AZ.

⑨ EC2 → compute service providing virtual servers to run applications

S3 → a ~~mag~~ managed object storage service for storing data.